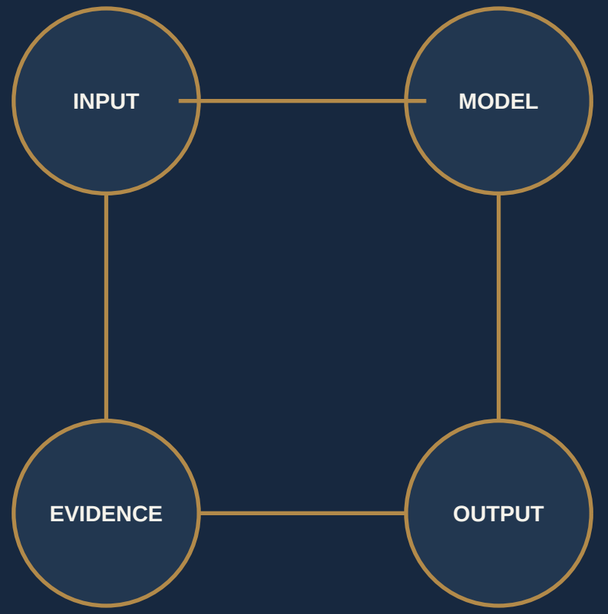
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Cognitive Robotics

AI.305 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define perception.
- Apply key methods including reasoning.
- Build or analyze a practical system involving memory.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Perception

02

METHODS

Reasoning

03

SYSTEMS

Memory

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Perception

Perception

02 · Reasoning

Reasoning

03 · Memory

Memory

04 · Learning

Learning

05 · Planning

Planning

06 · Decision-making and cognitive

Decision-making and cognitive architectures for robots

07 · Mandatory practical: Develop

Mandatory practical: Develop a cognitive robotic agent capable of perception

08 · Reasoning and action

Reasoning and action

Core ideas, made explicit

AI.305 • Distinguish definitions, assumptions and evidence

- Perception
- Reasoning
- Memory
- Learning
- Planning

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.305 • Distinguish definitions, assumptions and evidence

- Decision-making and cognitive architectures for robots
- Mandatory practical: Develop a cognitive robotic agent capable of perception
- Reasoning and action

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Perception

INPUTS
Reasoning

METHOD
Memory

OUTPUT
Learning

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving perception.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using perception.
- Implement or simulate a baseline using reasoning.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.